

A Process of Building an AI System

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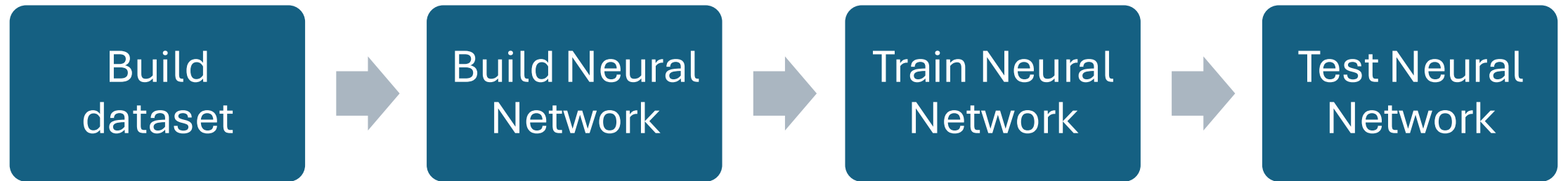
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Question

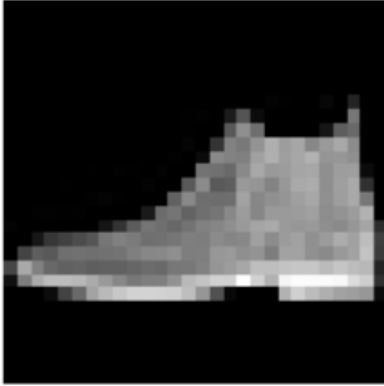
What do you think is the process for building an AI system?

Process for building AI systems



Build Dataset (I of II)

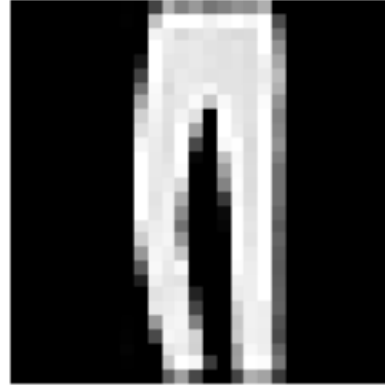
Ankle boot



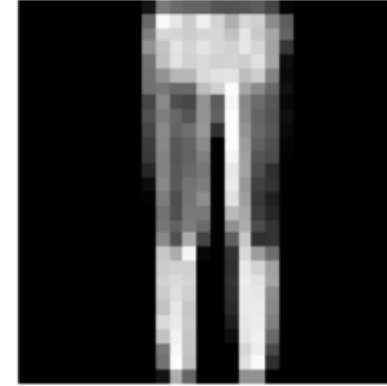
Pullover



Trouser



Trouser



You need to find a dataset that has a collection of images associated with labels:

Image1, “Ankle boot”

Image2, “Pullover”

Image3, “Trouser”

Image4, “Trouser”

:

Ideally, you want to have a **balanced dataset** with equal numbers of examples:

- 1,000 images of Ankle boots

- 1,000 images of Pullovers

- 1,000 Images of Trousers

The prompt “Where can I find a dataset for Neural Networks?” works for searching for datasets using a Chatbot.

Build Dataset (II of II)

Decide the number of images to be used for training and testing.
A typical choice is to use 80% for training and 20% for testing.

Thus, if we have 1,000 images, we will use:
80% of 1,000 = 800 images for training.
20% of 1,000 = 200 images for testing.

To select the images, we often use **random sampling with replacement**.

It is important that no images appear in both the training and testing sets.

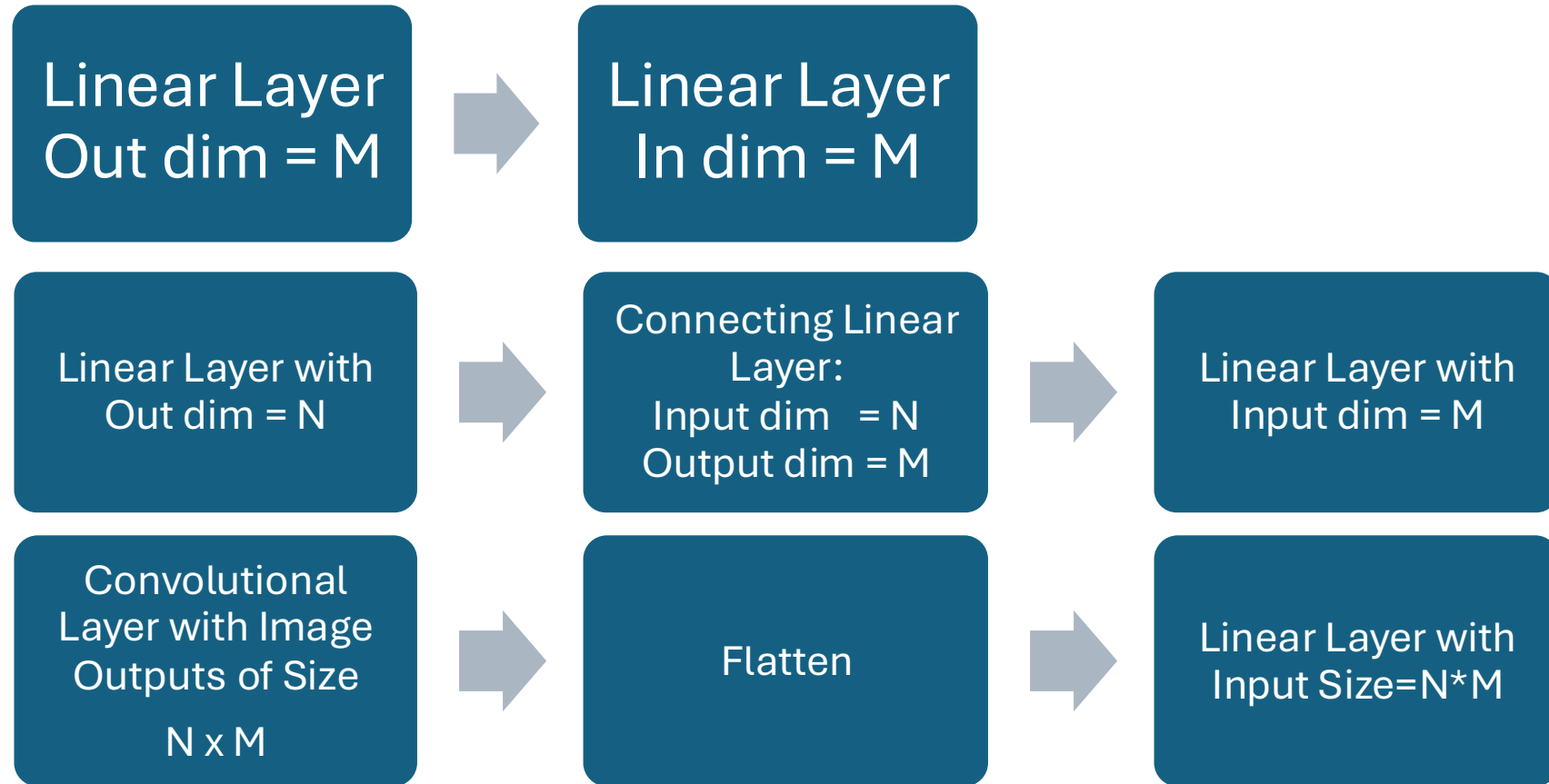
Why?

Also, the training and testing sets should have same distributions, ideally equal number for each category. Why? What could go wrong?

Build a Neural Network (I of II)

- Depending on the problem, determine the last layer:
 - Classification: you have one neuron per category with sigmoid activation functions.
 - Example: 10 neurons for 10 digits.
 - Regression: you have one neuron per output:
 - Example: 1 neuron for controlling the wheel of a car.
- Decide the image input size:
 - Example: 112x112
- Define layer types:
 - Convolutional layers with max pooling and ReLU activations are used for images.
 - Linear layers are used at the end.

Make sure your layers can connect



There are no restrictions for connecting two convolutional layers.

You need to figure out the dimensions in the code using `shape()` commands.

Training the Neural Network (I of II)

- Select the loss function based on the problem:
 - Use cross-entropy for classification problems.
- Select an optimization method using `torch.optim`.
- Select a learning rate or other parameters:
 - Look at recommendations by others or the official documentation

Example:

```
loss_fn = nn.CrossEntropyLoss()  
optimizer = torch.optim.SGD(model.parameters(), lr=1e-3)
```

Training the Neural Network (II of II)

```
1 epochs = 2 # Try 5, 10 but not too many!
2 for t in range(epochs):
3     print(f"Epoch {t+1}\n-----")
4     train(train_dataloader, model, loss_fn, optimizer)
5     test(test_dataloader, model, loss_fn)
6 print("Done!")
```

... Epoch 1

```
-----
loss: 2.302198 [ 64/60000]
loss: 2.279449 [ 6464/60000]
loss: 2.260805 [12864/60000]
loss: 2.269602 [19264/60000]
loss: 2.260987 [25664/60000]
loss: 2.246432 [32064/60000]
loss: 2.263424 [38464/60000]
loss: 2.228679 [44864/60000]
loss: 2.245714 [51264/60000]
loss: 2.258161 [57664/60000]
Test Error:
Accuracy: 28.6%, Avg loss: 2.208042
```

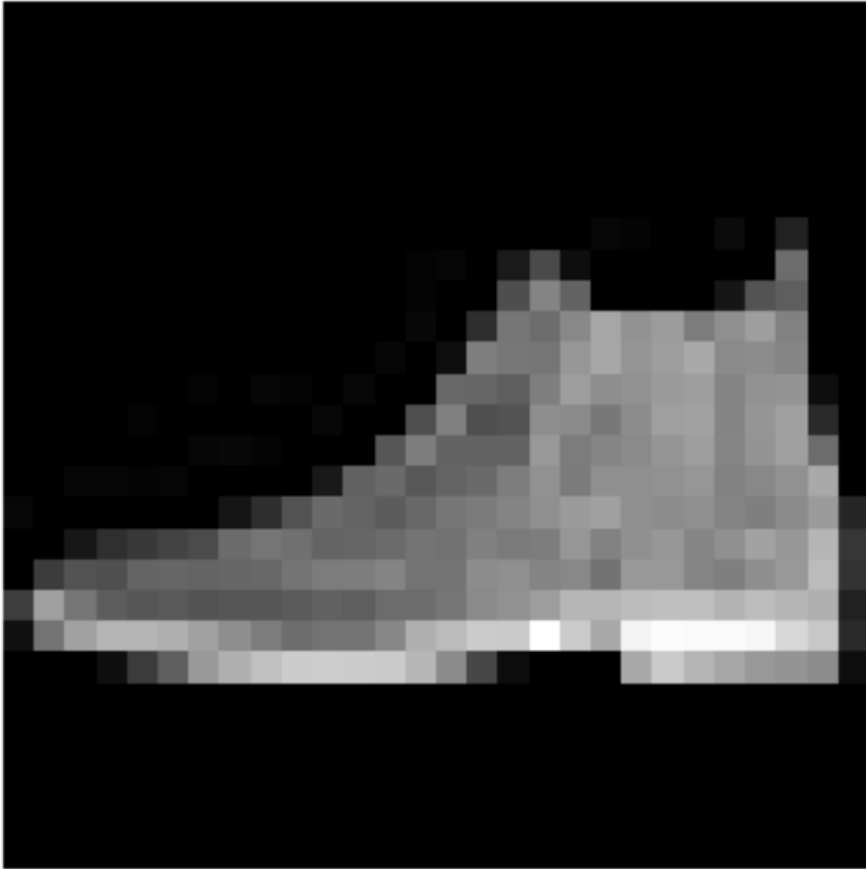
Start with a small number and increase the number of epochs.

You want the loss to decrease significantly and this should improve the accuracy.

What issues do you expect here?

Test the Neural Network

Predicted: "Ankle boot", Actual: "Ankle boot"



Try different input images and examine the outputs!

Compare predicted output labels versus the actual ones.

Determine when the network works and when it does not!